Why a Measure-Zero Periodic Set Can Contain Rational Parameters

Two elementary dynamical examples show why proving that periodic parameters are rare, measure zero, or isolated does not by itself exclude rational parameters.

1. A measure-zero periodic set containing every rational

Consider two uncoupled oscillators whose solution is

$$x(t) = \cos t, \qquad y(t) = \cos(ut), \qquad 0 < u < 1.$$

The complete labelled state returns precisely when the two frequencies are commensurable. Equivalently,

$$u \in \mathbb{Q}.$$

Thus the set of periodic parameters is

$$\mathbb{Q} \cap (0, 1).$$

This set is countable and has Lebesgue measure zero, yet it contains every rational parameter in the interval. It is also dense. Consequently, a theorem saying only that periodic parameters have measure zero would say nothing about whether the Pythagorean parameters are periodic.

2. An isolated periodic parameter that is rational

Consider

$$\ddot{x} + x = 0, \qquad \ddot{y} + 2y = 0,$$

with parameter-dependent initial conditions

$$x(0) = 1, \quad \dot{x}(0) = 0, \qquad y(0) = u - \frac{1}{3}, \quad \dot{y}(0) = 0.$$

The resulting solution is

$$x(t) = \cos t, \qquad y(t) = \left(u - \frac{1}{3}\right) \cos(\sqrt{2}\, t).$$

For $u \neq 1/3$, both incommensurable frequencies 1 and $\sqrt{2}$ are present, so the full orbit is quasiperiodic and never returns exactly. At

$$u = \frac{1}{3},$$

the second oscillator has zero amplitude, leaving the periodic motion $x(t) = \cos t$. Hence the periodic-parameter set is exactly

$$\left\{ \frac{1}{3} \right\},$$

an isolated, measure-zero set consisting of a rational number.

Relevance to the Pythagorean–Burrau conjecture

Showing that intersections with the periodic-brake locus are isolated would be substantial geometric progress, but it would not finish the conjecture. One would still need an exact bridge excluding rational intersection parameters—for example,

$$\mathcal{B}(u,t) = 0 \implies u \notin \mathbb{Q},$$

or the stronger real statement

$$\mathcal{B}(u,t) \neq 0 \qquad \text{for every real } u \in (0,1) \text{ and every admissible } t > 0.$$